# PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN CHARACTERISTIC 2

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Abstract. For normed spaces $E, E', F$ over a nontrivially normed field $k$ and a finite $\iota$, precomposition $f \mapsto (m \mapsto m \circ (f, \dots, f))$ sends continuous linear maps $E \to E'$ to continuous linear maps between spaces of continuous alternating maps. It is $C^n$ for every $n \in \mathbb{N} \cup \{\infty\}$ in every characteristic [5], and analytic when $|\iota|!$ is invertible in $k$ (by polarisation). We show that it is *not* analytic in general: over every complete ultrametric field of characteristic 2 that is not spherically complete, with $E = E' = \ell^\infty$ and $|\iota| = 2$, analyticity fails at 0. The proof is a finite parity identity for determinants in characteristic 2, glued along $\ell^\infty$; the failure is witnessed by two coordinate projections of $\ell^\infty$. The result is formalized in Lean 4/Mathlib in the form "the general $C^\omega$ statement implies `False`".

## 1. Introduction

Let $k$ be a nontrivially normed field. For normed spaces $E, E', F$ over $k$ and a finite $\iota$ we write $\mathcal{L}(E, E')$ for the space of continuous linear maps with the operator norm, and $\mathrm{Alt}^\iota(E, F)$ for the normed space of continuous alternating $\iota$-linear maps $E^\iota \to F$; we write $\mathrm{Alt}^2$ when $|\iota| = 2$.

**Analytic functors.** A map $f\colon X \to Z$ between normed spaces over $k$ is **analytic at** $x_0 \in X$ if there are bounded $n$-linear maps $p_n\colon X^n \to Z$ and a radius $r > 0$ with $\sup_n \|p_n\| r^n < \infty$ such that $f(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$ whenever $\|h\| < r$. It is **analytic**, or $C^\omega$, if it is analytic at every point of $X$; this is the definition used in Mathlib.

Lang [3, Ch. III, §4] calls a functor $T$ on Banach spaces **of class $C^p$** if $f \mapsto T(f)$ is $C^p$. Similarly, we call a contravariant endofunctor $T$ on the category of normed spaces over a field $k$ **analytic** if for all $E, E'$ the map

$$\mathcal{L}(E, E') \to \mathcal{L}\big(T(E'), T(E)\big), \qquad f \mapsto T(f),$$

is analytic. For a normed space $F$ and a finite $\iota$, the assignment $E \mapsto \mathrm{Alt}^\iota(E, F)$, $f \mapsto (m \mapsto m \circ (f, \dots, f))$, is a contravariant functor. The assignment is also functorial in $F$: a continuous linear map $g\colon F \to F'$ acts by $m \mapsto g \circ m$. This action is linear in $g$, hence analytic, whereas the action of $f$ is not linear in $f$. So the bifunctor $\mathrm{Alt}^\iota$ is analytic if and only if the functor $\mathrm{Alt}^\iota(-, F)$ is analytic for every $F$, and we fix $F$ from now on.

**Motivation: alternating bundles in Mathlib.** In Mathlib [4], manifolds and vector bundles are modelled on normed spaces (of possibly infinite dimension) over an arbitrary nontrivially normed field $k$, and their regularity is an element $n \in \mathbb{N} \cup \{\infty, \omega\}$, where $C^\omega$ means analytic. Differential forms require the bundle of continuous alternating maps between two $C^n$ vector bundles to be a $C^n$ vector bundle again (an instance of `ContMDiffVectorBundle`). Its transition functions are those of the given bundles with $\mathrm{Alt}^\iota$ applied, so this comes down to the regularity of the functor $\mathrm{Alt}^\iota(-, F)$, i.e. of the precomposition map

`ContinuousAlternatingMap.compContinuousLinearMapCLM`.

Two cases have been settled, in pull requests to Mathlib.

- If $|\iota|!$ is invertible in $k$ then $\mathrm{Alt}^\iota(-,F)$ is analytic, by polarisation. In Mathlib this is done in characteristic 0 by Gouëzel [1], who uses it to construct the bundle for every $n$ in characteristic 0 [2]; he remarks that a variant of the argument, not included there, covers finite-dimensional spaces in positive characteristic, but not the general case.
- Over every $k$, with no hypothesis on the characteristic or the dimension, $\mathrm{Alt}^\iota(-,F)$ is of class $C^n$ for every $n \in \mathbb{N}\cup\{\infty\}$ [5].

This leaves open the case where $|\iota|!$ is not invertible in $k$ and the spaces are infinite-dimensional: is $\mathrm{Alt}^\iota(-,F)$ analytic there? The following theorem answers this in the negative.

**Theorem 1.** *Let $k$ be a complete ultrametric field of characteristic 2 which is not spherically complete. There is a Banach space $F$ over $k$ such that the functor $\mathrm{Alt}^2(-,F)$ is not analytic.*

Such fields exist (Proposition 12). So $\mathrm{Alt}^\iota(-,F)$ is not analytic in general, already for $|\iota|=2$, and neither is the bifunctor $\mathrm{Alt}^\iota$. The terms used in the theorem are recalled in §2, where $F$ is given explicitly and the failure is located at $E = E' = \ell^\infty(\mathbb{N},k)$, the bounded sequences with the sup norm, and at the point $0 \in \mathcal{L}(E,E)$ (Theorem 2). Consequently the restriction $n \neq \omega$ in [5] cannot be removed, and a $C^\omega$ structure on bundles of continuous alternating maps needs a hypothesis such as the invertibility of $|\iota|!$. We do not treat the bundles themselves here, only the functor. In characteristic $\ell \geq 3$ the case $|\iota| \geq \ell$ remains open (Remark 15).

Theorems 1 and 2 and Proposition 12 are machine-checked in Lean 4 with Mathlib; together they give the statement verified there, "if precomposition is $C^\omega$ for all $k, E, E', F, \iota$, then `False`"; see §9. Sections 3–6 prove Theorem 2 in four steps.

## 2. Statement

**Notation.** For a polynomial $f$ and a monomial $\mu$, $\mathrm{coeff}_\mu f$ is the coefficient of $\mu$ in $f$. We write $c_0 \subset \ell^\infty$ for the null sequences, $\mathbf{1}_S$ for the indicator of a set $S$, and $\mathbf{1} = \mathbf{1}_{\mathbb{N}}$.

Throughout, $k$ is a field which is complete for a nontrivial absolute value $|\cdot|$ satisfying the **ultrametric inequality** $|x+y| \leq \max(|x|,|y|)$; we call such a field **ultrametric**. (Absolute values with this property are also called non-archimedean; this is a condition on the absolute value and is unrelated to non-Archimedean *ordered* fields.) The sup norms used below inherit the inequality. We assume that

(K1) $\mathrm{char}\, k = 2$;
(K2) $k$ is not spherically complete: there is a sequence of closed balls $B(c_n, r_n)$ such that $|c_{n+1} - c_n| \leq r_n$, $r_0 > r_1 > \cdots$ and $\bigcap_n B(c_n, r_n) = \emptyset$.

The balls in (K2) are balls in $k$, and they are nested by the ultrametric inequality: for $a \in B(c_{n+1}, r_{n+1})$ we have $|a - c_n| \leq \max(|a-c_{n+1}|, |c_{n+1}-c_n|) \leq \max(r_{n+1}, r_n) = r_n$. In characteristic 2 that inequality is in fact automatic: an absolute value satisfies it as soon as it is bounded on the multiples of 1, and here these are only 0 and 1. Such fields exist; an explicit one is given in §7. Let $E = \ell^\infty = \ell^\infty(\mathbb{N},k)$ with the sup norm, a Banach algebra under pointwise multiplication; write $m_a \in \mathcal{L}(E,E)$ for multiplication by $a \in E$, so that $\|m_a\| = \|a\|$ (evaluate at $\mathbf{1}$), and $e_i$ for the unit vectors. For $x, y \in E$ let

$$W(x,y) := x \otimes y + y \otimes x \in \ell^\infty(\mathbb{N}\times\mathbb{N}, k), \qquad W(x,y)_{pq} = x_p y_q + x_q y_p,$$

and let $F \subseteq \ell^\infty(\mathbb{N}\times\mathbb{N},k)$ be the closed linear span of all $W(x,y)$, a Banach space. Every element of $F$ is a symmetric matrix with zero diagonal (as $W(x,y)_{pp} = 2x_p y_p = 0$) and a uniform limit of matrices of finite rank (as $W(x,y)$ has rank $\leq 2$). By the ultrametric

inequality $\|W(x,y)\| \le \|x\|\|y\|$, and $W(x,x) = 2\,x \otimes x = 0$; so $W\colon E \times E \to F$ is a bounded alternating bilinear map of norm $\le 1$. Let

$$Q\colon \mathcal{L}(E,E) \to \mathcal{L}\big(\mathrm{Alt}^2(E,F), \mathrm{Alt}^2(E,F)\big), \qquad Q(f)(m) = m \circ (f,f).$$

**Theorem 2.** *$Q$ is not analytic at $0$. In particular $Q$ is not $C^\omega$.*

The map $Q$ is the action on $\mathcal{L}(E,E)$ of the functor $\mathrm{Alt}^2(-,F)$, so Theorem 2 implies Theorem 1.

## 3. Step 1: analyticity forces a bilinear lift

**Lemma 3.** *Let $Q\colon X \to Z$ be a map between normed spaces with $Q(tf) = t^2 Q(f)$. If $Q$ is analytic at $0$ then there is a bounded bilinear $P\colon X \times X \to Z$ with $P(f,f) = Q(f)$ for all $f$.*

*Proof.* Let $\sum_n p_n$ be a power series of $Q$ at $0$. For fixed $f$ the one-variable power series $\sum_n t^n\, p_n(f,\dots,f)$ and the polynomial $t^2 Q(f)$ agree for small $t$. Power series coefficients in one variable are unique over any nontrivially normed field, so $p_2(f,f) = Q(f)$. Take $P = p_2$. (No division by 2 occurs, and $P$ need not be symmetric.) □

So assume that a bounded bilinear $P$ with $P(f,f) = Q(f)$ exists, and put, for $a,b,x,y \in E$,

$$\Phi(a,b;x,y) := P(m_a, m_b)(W)(x,y) \in F.$$

Then $\Phi$ is 4-linear, and since $\|m_a\| = \|a\|$ and $\|W\| \le 1$ we have

$$\|\Phi(a,b;x,y)\| \le C\|a\|\,\|b\|\,\|x\|\,\|y\|, \qquad C := \max(1, \|P\|).$$

Moreover

$$\Phi(a,b;x,x) = 0, \qquad \Phi(a,a;x,y) = W(ax,ay). \tag{1}$$

Since $2 = 0$, the first identity gives the symmetry $\Phi(a,b;x,y) = \Phi(a,b;y,x)$. We will contradict the existence of such a $\Phi$.

## 4. Step 2: a finite parity identity

Fix a finite block $I \subset \mathbb{N}$ of even size $2m$ and $a', b' \in E$ with $\|a'\|, \|b'\| \le 1$ (in Step 4 they will carry the earlier blocks). Let $X_i, Y_i$ $(i \in I)$ be indeterminates. Expanding $\Phi$ multilinearly in $x = \sum_{i\in I} X_i e_i$ and $y = \sum_{i\in I} Y_i e_i$ defines polynomials

$$G_{pq} := \Phi(a'+x,\, b'+y;\, a'+x,\, b'+y)_{pq} \in k[X,Y] \qquad (p,q \in I),$$

whose values at points $x, y \in k^I$ (vectors supported on $I$) are the entries of an $I \times I$ matrix $G(x,y)$. The matrix $G$ is symmetric with zero diagonal, each $G_{pq}$ has degree $\le 2$ in $X$ and $\le 2$ in $Y$, its part of bidegree $(2,2)$ is $g_{pq} := \Phi(x,y;x,y)_{pq}$, and $G_{pq} - g_{pq}$ has total degree $\le 3$.

**Lemma 4.** *For $p \ne q$ we have, in $k[X,Y]$,*

$$g_{pq}(Y,X) = g_{pq}(X,Y) + X_p^2 Y_q^2 + X_q^2 Y_p^2 \qquad \text{and} \qquad g_{pq}(X,X) = 0.$$

*Proof.* A field with a nontrivial absolute value is infinite, so it suffices to check these at points $x, y \in k^I$. By the symmetry in the last two slots, $g(y,x) = \Phi(y,x;y,x) = \Phi(y,x;x,y)$, so by bilinearity in the first two slots and (1)

$$\begin{aligned} g(x,y) + g(y,x) &= \Phi(x+y, x+y; x, y) + \Phi(x,x;x,y) + \Phi(y,y;x,y) \\ &= W\big((x+y)x, (x+y)y\big) + W(x^2, xy) + W(xy, y^2). \end{aligned}$$

Expanding, $W((x+y)x, (x+y)y) = W(x^2,xy) + W(x^2,y^2) + W(xy,xy) + W(xy,y^2)$. The first and last terms cancel against the two other summands and $W(xy,xy) = 0$, which leaves $W(x^2,y^2)$, whose $(p,q)$ entry is $x_p^2 y_q^2 + x_q^2 y_p^2$. Finally $g(x,x) = W(x^2,x^2) = 0$. □

Let $\tau$ be the automorphism of $k[X,Y]$ swapping $X_i \leftrightarrow Y_i$, and $\delta$ the substitution $Y_i := X_i$.

**Lemma 5** (balanced coefficients)**.** *Let $f \in k[X,Y]$ with $\tau f = f$ and $\operatorname{char} k = 2$. For every $S \subseteq I$ the coefficient of $\prod_{i\in S} X_iY_i$ in $f$ equals the coefficient of $\prod_{i\in S} X_i^2$ in $\delta f$.*

*Proof.* The coefficient of $\prod_S X_i^2$ in $\delta f$ is the sum of the coefficients in $f$ of all monomials $X^\alpha Y^\beta$ with $\alpha+\beta = 2\cdot\mathbf{1}_S$. The involution $(\alpha,\beta)\mapsto(\beta,\alpha)$ of this set of monomials preserves coefficients (as $\tau f = f$), and its only fixed point is $\alpha=\beta=\mathbf{1}_S$; all other terms cancel in pairs. □

**Proposition 6.** *Let $M$ be a perfect matching of $I$ and $\nu := \prod_{i\in I} X_iY_i$. Then*

$$\operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq} = 1.$$

*Proof.* The monomial $\nu$ has total degree $4m$, the maximum possible, so only the top parts contribute:

$$\operatorname{coeff}_\nu \prod_M G_{pq} = \operatorname{coeff}_\nu \prod_M g_{pq}.$$

Put $g'_{pq} := g_{pq} + X_p^2Y_q^2$. By Lemma 4,

$$\tau g'_{pq} = g'_{pq} \qquad \text{and} \qquad \delta g'_{pq} = X_p^2X_q^2.$$

A product containing a factor $X_p^2Y_q^2$ has no squarefree monomial, so replacing every $g_{pq}$ by $g'_{pq}$ does not change the coefficient of $\nu$. By Lemma 5, applied to the $\tau$-invariant product $\prod_M g'_{pq}$, this coefficient is the coefficient of $\prod_{i\in I} X_i^2$ in

$$\delta \prod_M g'_{pq} = \prod_{\{p,q\}\in M} X_p^2X_q^2 = \prod_{i\in I} X_i^2,$$

namely 1. □

**Lemma 7.** *Let $A$ be a symmetric matrix with zero diagonal over a commutative ring of characteristic 2, of even size $2m$. Then $\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2$, the sum being over perfect matchings; and the number of perfect matchings, $(2m-1)!!$, is odd.*

*Proof.* In $\det A = \sum_\sigma \prod_i A_{i\sigma(i)}$ (all signs are 1), $\sigma$ and $\sigma^{-1}$ contribute equally by symmetry, so only involutions survive; those with a fixed point contribute 0; a fixed-point-free involution is a perfect matching and contributes $\prod A_{pq}^2$. □

**Lemma 8** (hypercube sum)**.** *Let $R$ be a commutative ring of characteristic 2 and let $f \in R[Z_1,\dots,Z_N]$ have total degree $\le N$. Then $\sum_{z\in\{0,1\}^N} f(z)$ is the coefficient of $Z_1Z_2\cdots Z_N$ in $f$.*

*Proof.* By linearity we may take $f$ to be a monomial $Z^\alpha$. At a point $z\in\{0,1\}^N$ it takes the value 1 if $z_i=1$ for all $i$ in the support of $\alpha$, and 0 otherwise; so the sum is $2^{N-|\operatorname{supp}\alpha|}$, which vanishes in $R$ unless every variable occurs in $Z^\alpha$. The only monomial of degree $\le N$ in which all $N$ variables occur is $Z_1\cdots Z_N$. □

**Corollary 9.** *The values of $\det G$ at the $0/1$ points sum to 1,*

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = 1,$$

*and consequently there are $x,y\in\{0,1\}^I$ with $|\det G(x,y)| \ge 1$.*

*Proof.* For a perfect matching $M$ the product $\prod_{\{p,q\}\in M} G_{pq}$ has total degree $\le 4m$, which is the number of variables $X_i, Y_i$. By Lemma 8 and Proposition 6 its sum over the 0/1 points is therefore $\operatorname{coeff}_\nu \prod_M G_{pq} = 1$. Squaring is additive in characteristic 2, so by Lemma 7

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = \sum_M \Big( \sum_{x,y\in\{0,1\}^I} \prod_{\{p,q\}\in M} G_{pq}(x,y) \Big)^2 = (2m-1)!! \cdot 1 = 1,$$

because $(2m-1)!!$ is odd. If all the values $\det G(x,y)$ had absolute value $< 1$, the ultrametric inequality would give $|1| < 1$. □

The point is that the coefficient 1 does not depend on the unknown lift $P$: all of its freedom is carried by $\tau$-invariant terms vanishing on the diagonal $Y = X$, which Lemma 5 cannot see. No single pair $(x, y)$ works for every lift, but the $2^{4m}$ vertices of the cube cannot all fail at once.

## 5. Step 3: locality on $\ell^\infty$

For $x \in \ell^\infty$ write $x|_{\ge N}$ for $x$ with its coordinates below $N$ set to 0.

**Proposition 10.** *Under* (K2)*, a bounded linear functional $\varphi$ on $\ell^\infty$ with $\varphi(e_i) = 0$ for all $i$ is zero.*

*Proof.* $\varphi$ kills finitely supported sequences, so $|\varphi(x)| \le \|\varphi\| \sup_{i\ge M} |x_i|$ for every $M$. Suppose $\varphi \ne 0$. For every $0 < s < \|\varphi\|$ there is $w$ with $\varphi(w) = 1$ and $\|w\| \le 1/s$; fix one such vector $w_0$. Because $r_{n+1} < r_n$ we can choose $s_n \in \ell^\infty$ with $\varphi(s_n) = c_{n+2} - c_{n+1}$ and $\|s_n\| \le r_n/\|\varphi\|$: take $s_n = (c_{n+2} - c_{n+1})w$ with $\|w\| \le r_n/(r_{n+1}\|\varphi\|)$. Let $x_n := c_1 w_0 + \sum_{k<n} s_k$; then $\varphi(x_n) = c_{n+1}$ and $\|x_{n+d} - x_n\| \le r_n/\|\varphi\|$ by the ultrametric inequality. The diagonal sequence $y_i := x_i(i)$ is bounded, since $\|x_i\| \le \max(\|x_0\|, r_0/\|\varphi\|)$, and $|y_i - x_n(i)| \le r_n/\|\varphi\|$ for $i \ge n$. Hence $|\varphi(y) - c_{n+1}| = |\varphi(y - x_n)| \le r_n$, and since $|c_{n+1} - c_n| \le r_n$ the ultrametric inequality gives $|\varphi(y) - c_n| \le r_n$ for every $n$. So $\varphi(y) \in \bigcap_n B(c_n, r_n)$, a contradiction. □

This is the direction of van der Put's theorem [6] that we need, $(\ell^\infty/c_0)' = 0$; the index shift absorbs the usual $(1+\varepsilon)$ loss, and no Hahn–Banach argument is used.

**Proposition 11** (multilinear Phillips property)**.** *Let $\lambda$ be a bounded $d$-linear form on $\ell^\infty$ and $\varepsilon > 0$. There is $N$ such that $|\lambda(u_1, \dots, u_d)| \le \varepsilon$ whenever $\|u_j\| \le 1$ for all $j$ and $u_j = u_j|_{\ge N}$ for some $j$.*

*Sketch* (the complete proof is in the Lean development). *The case $d = 1$, $\lambda = \varphi$.* (a) $\varphi(e_i) \to 0$. Otherwise $|\varphi(e_{i_k})| \ge \varepsilon$ along a subsequence. Let $T$ be the bounded operator with $T(x)_{i_k} = x_k/\varphi(e_{i_k})$ and all other coordinates 0, and $\psi := \varphi \circ T$, so that $\psi(e_k) = 1$ for all $k$. With sh the right shift, $\psi \circ (1 - \mathrm{sh})$ kills every $e_k$, hence vanishes by Proposition 10; but $(1 - \mathrm{sh})\mathbf{1} = e_0$, so $0 = \psi(e_0) = 1$. (b) For fixed $x$, $\varphi(x|_{\ge N}) - \varphi(x|_{\ge N+1}) = x_N \varphi(e_N) \to 0$, so $\varphi(x|_{\ge N})$ converges; the limit is a bounded linear functional of $x$ killing every $e_i$, hence it is 0. (c) If the claim failed for $\varepsilon$, then (b) would give unit vectors $w_k$ with disjoint finite supports moving to infinity and $|\varphi(w_k)| > \varepsilon$; the functional $c \mapsto \varphi(\sum_k c_k w_k)$ (coordinatewise sum) then contradicts (a).

*The case $d > 1$*, by induction on $d$. Suppose the claim fails for $\varepsilon$. First, for every $M$ there is a unit tuple with $|\lambda| > \varepsilon$ *all* of whose slots vanish below $M$: start from a witness one of whose slots is supported very far out, and split any other slot into its part below $M$, a finite combination of the $e_i$, and its tail; the inductive hypothesis, applied to the finitely many $(d-1)$-linear forms obtained by freezing that slot at $e_i$, $i < M$, shows that the part below $M$ contributes at most $\varepsilon$, so the tail carries $|\lambda| > \varepsilon$. Next, by (b) each slot can be cut down to a finite window. Finally choose such tuples $z^{(s)}$ with windows moving to infinity fast enough and

let $x$ be the coordinatewise sum of their first slots. Then $\lambda(x, z_2^{(s)}, \dots, z_d^{(s)})$ equals $\lambda(z^{(s)})$, plus finitely many terms with an earlier first slot (each $\le \varepsilon$ by the inductive hypothesis for $\lambda(z_1^{(r)}, \cdot)$, $r < s$), plus one term with a far first slot ($\le \varepsilon$ by the case $d = 1$). So it has absolute value $> \varepsilon$ for every $s$, contradicting the inductive hypothesis for the $(d-1)$-linear form $\lambda(x, \cdot)$. □

This is exactly where $\ell^\infty$ and (K2) are used; the statement is false for $c_0$ and for spherically complete fields.

## 6. Step 4: gluing

Choose inductively blocks $I_1 < I_2 < \cdots$ with $|I_t| = 2t$ and vectors $a_t, b_t \in \{0,1\}^{I_t}$ as follows. Given the earlier stages let $a' = \sum_{s<t} a_s$, $b' = \sum_{s<t} b_s$, and apply Corollary 9 to get $a_t, b_t$ with $|\det G_t| \ge 1$, where $G_t = \Phi(a' + a_t, b' + b_t; a' + a_t, b' + b_t)|_{I_t \times I_t}$. Put $\varepsilon_t := \frac{1}{2} C^{-2t}$ and use Proposition 11 (with $d = 4$) for the finitely many entry forms $\Phi(\cdot)_{pq}$, $p, q \in I_t$, to place all later blocks so far out that these forms are at most $\varepsilon_t$ on unit arguments one of which is supported there.

Let $a, b \in \ell^\infty$ be defined by $a|_{I_t} = a_t$, $b|_{I_t} = b_t$ and zero elsewhere. These are not norm-convergent sums; $a = \mathbf{1}_A$ and $b = \mathbf{1}_B$ are indicator functions of two subsets of $\mathbb{N}$, so $m_a, m_b$ are coordinate projections. Let $\mathcal{O} := \Phi(a, b; a, b) \in F$. Splitting $a = a^{\le t} + a^{>t}$, $b = b^{\le t} + b^{>t}$ and using multilinearity, $\|\mathcal{O}|_{I_t \times I_t} - G_t\| \le \varepsilon_t$. All entries are bounded by $C$. Expanding the determinant multilinearly in columns, every term containing a column of the perturbation is at most $\varepsilon_t C^{2t-1} < 1$, so the ultrametric inequality gives

$$|\det \mathcal{O}|_{I_t \times I_t}| = |\det G_t| \ge 1 \qquad \text{for every } t.$$

On the other hand $\mathcal{O} \in F$ is within $\eta := (2C)^{-1}$ of a matrix $R$ of some finite rank $r$. Expanding $\det(R + (\mathcal{O} - R))$ on $I_t \times I_t$ in the same way, terms with more than $r$ columns from $R$ vanish, so $|\det \mathcal{O}|_{I_t \times I_t}| \le C^r \eta^{2t-r} < 1$ once $t > r$. This contradiction proves Theorem 2. □

## 7. An explicit field

The purpose of this section is to construct a field satisfying (K1) and (K2). Towards that end, let $k_0 = \mathbb{F}_2$ (or any field of characteristic 2). For $j \ge 0$ let $k_0((t^{1/2^j}))$ be the field of *formal* Laurent series $\sum_{n \ge N} a_n t^{n/2^j}$ ($N \in \mathbb{Z}$, $a_n \in k_0$) in the single variable $t^{1/2^j}$. These fields form an increasing chain, and we let

$$\mathcal{P} := \bigcup_{j \ge 0} k_0((t^{1/2^j})), \qquad |f| := 2^{-\operatorname{ord} f},$$

be the field of **2-power Puiseux series**. Here $\operatorname{ord} f$ is the minimum $q$ for which $\operatorname{coeff}_{t^q} f \ne 0$. We set $|0| = 0$ by definition. Let $k$ be the completion of $\mathcal{P}$ for this (ultrametric) absolute value. This is characteristic 2 since $k_0 \hookrightarrow \mathcal{P} \hookrightarrow k$.

**Proposition 12.** *$k$ is not spherically complete.*

*Proof.* We exhibit a sequence of balls as in (K2). Put

$$\gamma_m := 1 - 2^{-(m+1)}, \qquad c_n := \sum_{m=0}^{n} t^{\gamma_m} \in \mathcal{P}, \qquad r_n := 2^{-\gamma_{n+1}}.$$

*The balls have the form required in* (K2). Since $c_{n+1} - c_n = t^{\gamma_{n+1}}$ we have $|c_{n+1} - c_n| = 2^{-\gamma_{n+1}} = r_n$. Since the $\gamma_m$ are strictly increasing to 1, the $r_n$ are strictly decreasing, and $r_n > 2^{-1} = \frac{1}{2}$.

*The intersection is empty.* Suppose that $a \in k$ satisfies $|a - c_n| \le r_n$ for all $n$. As $\mathcal{P}$ is dense in $k$ there is $y \in \mathcal{P}$ with $|a - y| < \frac{1}{2}$, and then for every $n$

$$|y - c_n| \le \max\big(|y - a|, |a - c_n|\big) \le \max\big(\tfrac{1}{2}, r_n\big) = r_n. \tag{2}$$

Now $y \in k_0((t^{1/2^j}))$ for some $j$. The exponent $\gamma_j = (2^{j+1} - 1)/2^{j+1}$ has an odd numerator, therefore $\gamma_j$ is not a multiple of $1/2^j$ and so $\operatorname{coeff}_{t^{\gamma_j}} y = 0$. For $n \ge j$ we have $2^{-\gamma_j} > 2^{-\gamma_{n+1}} = r_n$. Furthermore, the term $t^{\gamma_j}$ occurs in $c_n$, so

$$\operatorname{coeff}_{t^{\gamma_j}}(y - c_n) = 0 - 1 \neq 0, \qquad \text{hence} \qquad \operatorname{ord}(y - c_n) \le \gamma_j \quad \text{and} \quad |y - c_n| \ge 2^{-\gamma_j} > r_n,$$

which contradicts (2). □

## 8. Remarks

*Remark* 13 (Why characteristic 2, why $\ell^\infty$, why this $F$). In characteristic $\neq 2$ the lift $\frac{1}{2}\big(m \circ (f, g) + m \circ (g, f)\big)$ exists, and the parity argument breaks (pairs add instead of cancelling). If $E$ has an orthogonal base $(e_i)$, e.g. $E = c_0$, and $F$ is Banach, then

$$P(f, g)(m)(x, y) = \sum_{i<j} (x_i y_j - x_j y_i)\, m(f e_i, g e_j)$$

is a bounded lift in every characteristic, so $Q$ is analytic; a similar formula works for a base of $E'$, and $t$-orthogonal bases suffice. This covers finite-dimensional $E$ and all Banach spaces over a discretely valued $k$ [7], so the counterexample needs a dense valuation; one can also show that $Q$ is analytic when $F$ is spherically complete. On $c_0$ the glued vectors $a, b$ of Step 4 do not exist and Proposition 11 fails. For $E = \ell^\infty$ and the larger target $\ell^\infty(\mathbb{N} \times \mathbb{N}, k) \supseteq F$ a bounded lift also exists (sort the indices entrywise), but it sends $W$ to matrices such as the "staircase" $\mathbf{1}_{\{(p,q)\,:\,p \text{ odd}, q \text{ even}, p<q\}}$, which are not limits of finite-rank matrices, so it leaves $F$. These positive statements are not proved here and are not part of the formalization.

*Remark* 14 (What is forced). No single pair $(x, y)$ works for every lift: adding to $\Phi$ a bounded 4-linear map with values in $F$ that is alternating in $(a, b)$ and in $(x, y)$ gives another admissible $\Phi$, and this can change a given off-diagonal entry of $G(x, y)$ by an arbitrary scalar. Blocks of bounded size cannot give a contradiction either: for $u = \mathbf{1}_{\text{odd}}$, $v = \mathbf{1}_{\text{even}}$ the rank-two matrix $W(u, v) \in F$ has the entry 1 in every diagonal block $\{2s - 1, 2s\}$. On a finite block the sorted lift of the previous remark has norm 1 and satisfies all the identities of Step 2, so there is no finite-dimensional obstruction: the invariant is the rank of one element of $F$ against the size of the blocks, and Step 3 is what allows infinitely many blocks to be glued into one element.

*Remark* 15 (Higher degree, other characteristics). On paper, in characteristic 2 every $|\iota| \ge 2$ is covered by wedging with a coordinate of $E \oplus k$; this is not formalized. In odd characteristic $\ell$, for forms of degree $\ell$, the question is open: odd-degree forms have no Pfaffian-type top power, so a different invariant is needed.

## 9. Formalization

The Lean 4/Mathlib development is at

https://github.com/Deicyde/alternating-analytic-counterexample

The library `AltCounter/` contains no `sorry`, and the only axioms it uses are the three standard ones (`propext`, `Quot.sound` and `Classical.choice`). Its headline theorem assumes that Mathlib's

`ContinuousAlternatingMap.compContinuousLinearMapCLM`

is $C^\omega$ for all nontrivially normed fields, normed spaces and finite index types, and derives `False`; a second theorem does the same assuming this only for complete ultrametric fields of characteristic 2, Banach spaces, $E' = E$ and $|\iota| = 2$. Both are restated in a Mathlib-only file `Challenge.lean` (with `sorry` in place of the proofs, as that format requires) and checked against `Solution.lean` by the `leanprover/comparator` tool. Steps 1–4 and §7 are formalized, including the complete proof of Proposition 11 by induction on $d$; Lemma 8 is `sum_eval_cubePt`, and the field of §7 is built with $k_0 = \mathbb{F}_2$. The Lean constants in Step 4 are chosen with more slack than here, and the Lean form of Corollary 9 records only that the witnesses lie in the unit ball, which is all that Step 4 uses. Not formalized: non-analyticity at base points other than 0, index types with more than two elements, the positive statements in the remarks, and the bundle-level consequence (no $C^\omega$ vector-bundle structure on bundles of continuous alternating maps in this generality).

## References


[1] S. Gouëzel, *In characteristic* 0*, precomposition by a linear map is analytic on the space of alternating maps*, pull request #43338 to Mathlib, https://github.com/leanprover-community/mathlib4/pull/43338.

[2] S. Gouëzel, *The bundle of continuous alternating maps is* $C^n$, draft pull request #43234 to Mathlib, https://github.com/leanprover-community/mathlib4/pull/43234.

[3] S. Lang, *Fundamentals of Differential Geometry*, Graduate Texts in Mathematics 191, Springer, 1999.

[4] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 367–381.

[5] J. McCarthy, *Smoothness reflects along a linear isometry with closed range*, draft pull request #43548 to Mathlib, https://github.com/leanprover-community/mathlib4/pull/43548.

[6] M. van der Put, *Espaces de Banach non archimédiens*, Bull. Soc. Math. France **97** (1969), 309–320.

[7] A. C. M. van Rooij, *Non-Archimedean Functional Analysis*, Marcel Dekker, 1978.